

DATA

EXPLORATION

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EXPLORATION

The following concepts will be introduced:

✓ SUMMARY STATISTICS

- MEAN, MODE
- QUANTILE/PERCENTILE
- RANGE, VARIANCE, STANDARD DEVIATION
- AAD, MAD, IQR

✓ VISUALIZATION

- HISTOGRAM
- BOX-AND-WHISKERS

EXPLORATION: SUMMARY STATISTICS

1

After two days you receive an **EMAIL MESSAGE FROM YOUR FRIEND** with attached a txt file named churn.



You download and open the **CHURN.TXT FILE AND INSPECT THE FIRST 10 LINES:**

Area Code	Day Mins	Eve Mins	Churn	Int'l Plan	VMail Plan	Day Calls	Night Calls	Night Charge	Intl Calls	State	Phone
415	265.1	197.4	n	0	1	110	91	11.01	3	?	382-4657
415	161.6	195.5	n	0	1	123	103	11.45	3	OH	371-7191
?	243.4	121.2	n	0	0	114	104	7.32	5	NJ	358-1921
408	299.4	61.9	n	?	0	71	89	8.86	7	OH	375-9999
415	166.7	148.3	y	1	0	113	121	8.41	3	OK	330-6626
510	223.4	220.6	n	1	0	98	118	9.18	?	AL	?
510	218.2	348.5	n	0	1	88	118	9.57	7	MA	355-9993
415	157	103.1	n	1	0	?	96	9.53	6	MO	329-9001
408	184.5	351.6	n	0	0	97	90	9.71	4	LA	335-4719
415	?	222	n	1	1	84	97	14.69	5	WV	330-8173

Your friend asks you to provide **THE MOST BASIC SUMMARY OF THE DATA** to gain a general picture of **HOW CHURNING IS PROGRESSING**.

You make the decision to first compute **SUMMARY STATISTICS** for all the **ATTRIBUTES** reported in the churn.txt data file, when this is meaningful.

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The **CHURN ATTRIBUTE IS QUALITATIVE** and so you **COMPUTE** it's **MODE**, i.e., the **MOST FREQUENT VALUE IN THE DATA SET:**

	n	y	
absolute frequency	9	1	10
	9/10	1/10	
relative frequency	0.9	0.1	1

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415	?	222	n	1	1	84	97	14.69	5	WV	330-8173

The **DAY MINS ATTRIBUTE IS QUANTITATIVE** and so you consider a different summary statistic, the **QUANTILES** of a set of values.

157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4
265.1	161.6	243.4	299.4	166.7	223.4	218.2	157.0	184.5

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quantile of order
1/3



157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4
265.1	161.6	243.4	299.4	166.7	223.4	218.2	157.0	184.5

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quantile of order
2/3



157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4
265.1	161.6	243.4	299.4	166.7	223.4	218.2	157.0	184.5

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MEDIAN



157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4
265.1	161.6	243.4	299.4	166.7	223.4	218.2	157.0	184.5

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You also compute the **MEAN OF DAY MINS.**

$$mean = \frac{1}{9} \cdot \sum_{i=1}^9 x_i = 213.26$$

x_1	x_2	...	...	...	...	...	x_8	x_9
265.1	161.6	243.4	299.4	166.7	223.4	218.2	157.0	184.5

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The **MEAN** is sensitive to anomalous records (**OUTLIERS**) while the **MEDIAN** is a **MORE ROBUST ESTIMATE OF THE MIDDLE** of a set of values.

$$mean = \frac{1}{9} \cdot \sum_{i=1}^9 x_i = 213.26$$

$X_{(1)}$	$X_{(2)}$	...	...	...	...	...	$X_{(8)}$	$X_{(9)}$
157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4

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The **TRIMMED MEAN** is sometimes used.

$$mean = \frac{1}{9} \cdot \sum_{i=1}^9 x_i = 213.26$$

$X_{(1)}$	$X_{(2)}$	...	...	...	...	...	$X_{(8)}$	$X_{(9)}$
157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4

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The **TRIMMED MEAN** is sometimes used.

$$mean = \frac{1}{9} \cdot \sum_{i=1}^9 x_i = 213.26$$

$x_{(2)}$ $x_{(8)}$
161.6 166.7 184.5 218.2 223.4 243.4 265.1

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$$mean = \frac{1}{9} \cdot \sum_{i=1}^9 x_i = 213.26 \quad \boxed{208.98}$$

$x_{(2)}$ $x_{(8)}$
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Another set of used **summary statistics for quantitative attributes** are those that **measure the dispersion (spread)** of a set of values. **RANGE**

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Another set of used summary statistics for quantitative attributes are those that measure the dispersion (spread) of a set of values. **RANGE**

$X_{(1)}$	$X_{(2)}$	...	...	...	...	...	$X_{(8)}$	$X_{(9)}$
157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4

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Another set of used summary statistics for quantitative attributes are those that measure the dispersion (spread) of a set of values. **RANGE**

MIN
↓

$X_{(1)}$

157.0

$X_{(2)}$

161.6

...

166.7

...

184.5

...

218.2

...

223.4

...

243.4

$X_{(8)}$

265.1

MAX
↓

$X_{(9)}$

299.4

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MIN



$X_{(1)}$

157.0

$X_{(2)}$

161.6

...

166.7

...

184.5

...

218.2

...

223.4

...

243.4

$X_{(8)}$

265.1

MAX



$X_{(9)}$

299.4

$$range = 299.4 - 157.0 = 142.4$$

EXPLORATION: SUMMARY STATISTICS

1

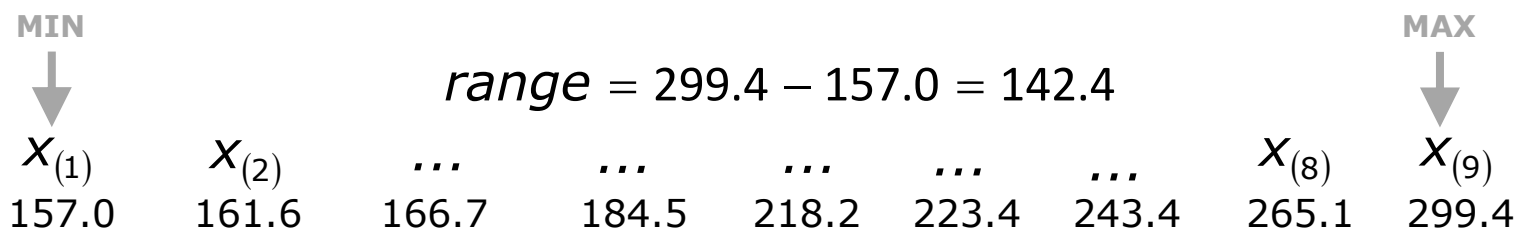
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415	?	222	n	1	1	84	97	14.69	5	WV	330-8173

RANGE, can be misleading if most of the values are concentrated in a narrow band of values.



EXPLORATION: SUMMARY STATISTICS

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415	?	222	n	1	1	84	97	14.69	5	WV	330-8173

RANGE, can be misleading if most of the values are concentrated in a narrow band of values.

The **VARIANCE** is preferred.

$$var = \frac{1}{8} \cdot \sum_{i=1}^9 (x_i - 213.26)^2 = 2,496.5$$

$X_{(1)}$	$X_{(2)}$	...	...	...	...	...	$X_{(8)}$	$X_{(9)}$
157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4

EXPLORATION: SUMMARY STATISTICS

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RANGE, can be misleading if most of the values are concentrated in a narrow band of values.

The **VARIANCE** is preferred.

STANDARD DEVIATION

$$var = \frac{1}{8} \cdot \sum_{i=1}^9 (x_i - 213.26)^2 = 2,496.5$$

$$\sigma = \sqrt{var} = 49.96$$

$X_{(1)}$	$X_{(2)}$	...	...	...	...	...	$X_{(8)}$	$X_{(9)}$
157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4

EXPLORATION: SUMMARY STATISTICS

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415	?	222	n	1	1	84	97	14.69	5	WV	330-8173

The variance depends on the mean and thus it is also sensitive to outliers. More robust estimates of the spread of a set of values are **ABSOLUTE AVERAGE DEVIATION**

$$AAD = \frac{1}{9} \cdot \sum_{i=1}^9 |x_i - 213.26| = 40.72$$

$x_{(1)}$	$x_{(2)}$	...	...	...	...	...	$x_{(8)}$	$x_{(9)}$
157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4

EXPLORATION: SUMMARY STATISTICS

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The variance depends on the mean and thus it is also sensitive to outliers. More robust estimates of the spread of a set of values are **MEDIAN ABSOLUTE DEVIATION**

$$MAD = median(|x_{(1)} - 213.26|, \dots, |x_{(9)} - 213.26|) = 46.56$$

$x_{(1)}$	$x_{(2)}$	...	...	...	...	...	$x_{(8)}$	$x_{(9)}$
157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4

EXPLORATION: SUMMARY STATISTICS

1

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The variance depends on the mean and thus it is also sensitive to outliers. More robust estimates of the spread of a set of values are **INTERQUARTILE RANGE (IQR)**

$X_{(1)}$	$X_{(2)}$	...	...	...	...	...	$X_{(8)}$	$X_{(9)}$
157.0	161.6	166.7	184.5	218.2	223.4	243.4	265.1	299.4

EXPLORATION: SUMMARY STATISTICS

1

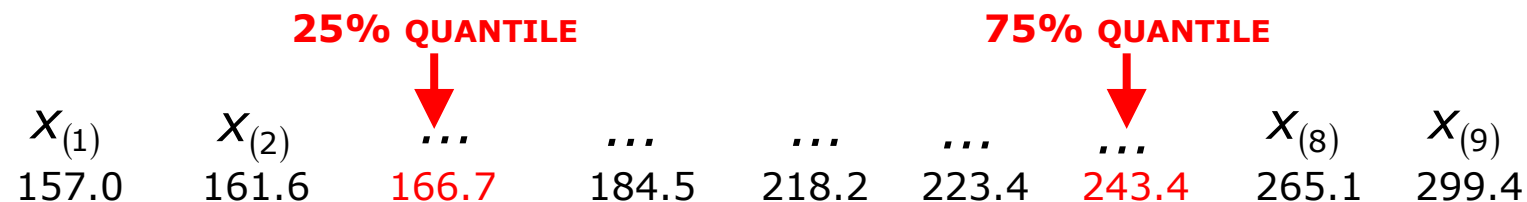
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EXPLORATION: SUMMARY STATISTICS

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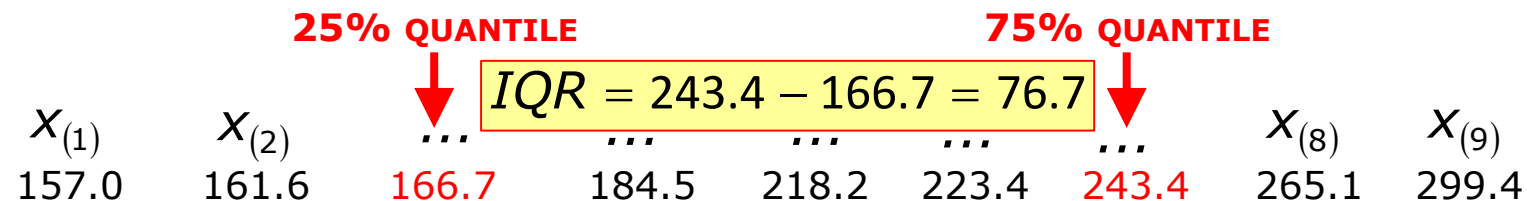
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When **multiple Quantitative Attributes** are available you usually compute the **VARIANCE-COVARIANCE MATRIX**

$$\text{cov}(X, Y) = \frac{1}{m-1} \cdot \sum_{i=1}^m (x_i - \bar{x}) \cdot (y_i - \bar{y})$$

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The **VARIANCE-COVARIANCE MATRIX** is square, symmetric and its “ ij ” element is the covariance between the i^{th} attribute and the j^{th} attribute.

EXPLORATION: SUMMARY STATISTICS

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Another **measure of association between pairs of Quantitative Attributes** which does not depend on the variance of each attribute is the **LINEAR CORRELATION COEFFICIENT**

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(x) \cdot \text{var}(y)}}$$

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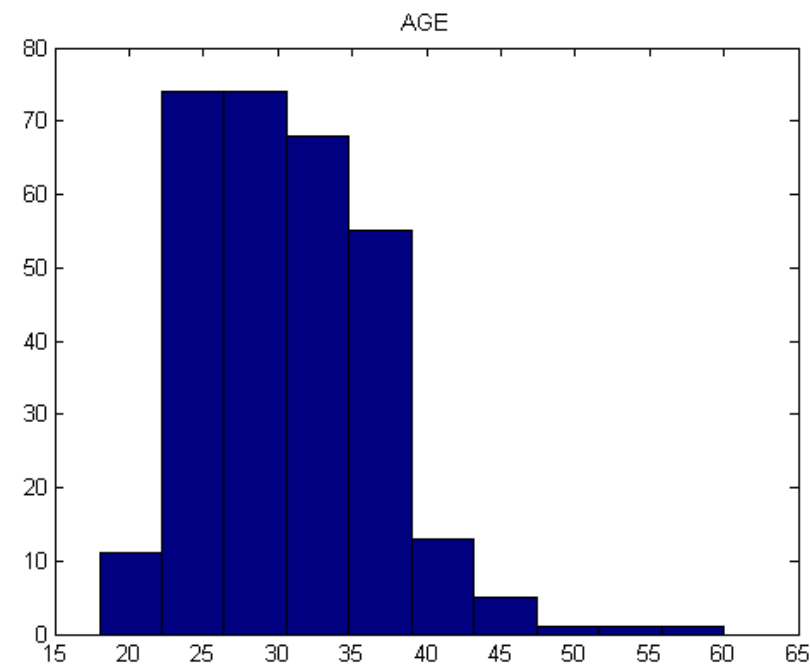
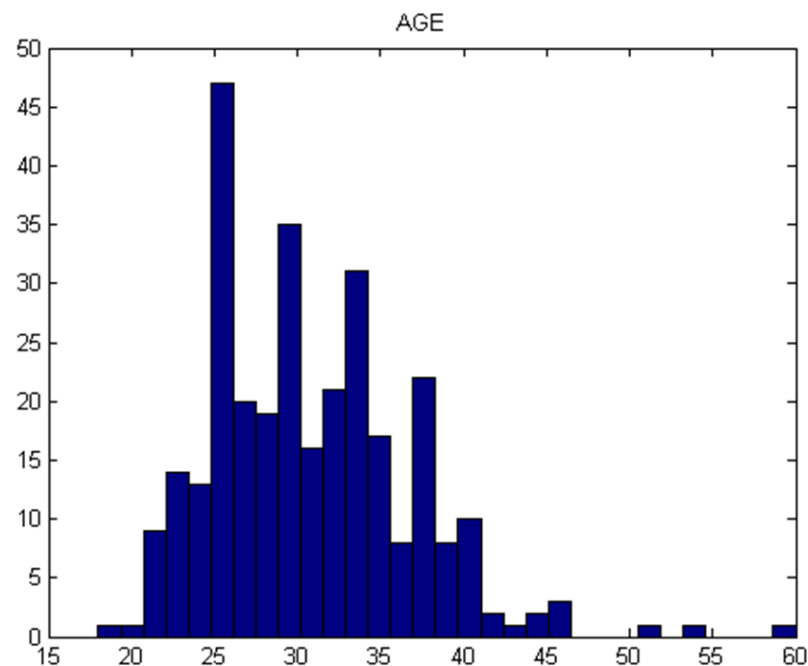
$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(x) \cdot \text{var}(y)}}$$

The **LINEAR CORRELATION COEFFICIENT** ranges in $[-1, +1]$, the greater it is in absolute value the stronger it is the linear relationship between the two attributes.

You also make the decision to exploit **DATA VISUALIZATION**, i.e., to display information in a graphic or tabular format.

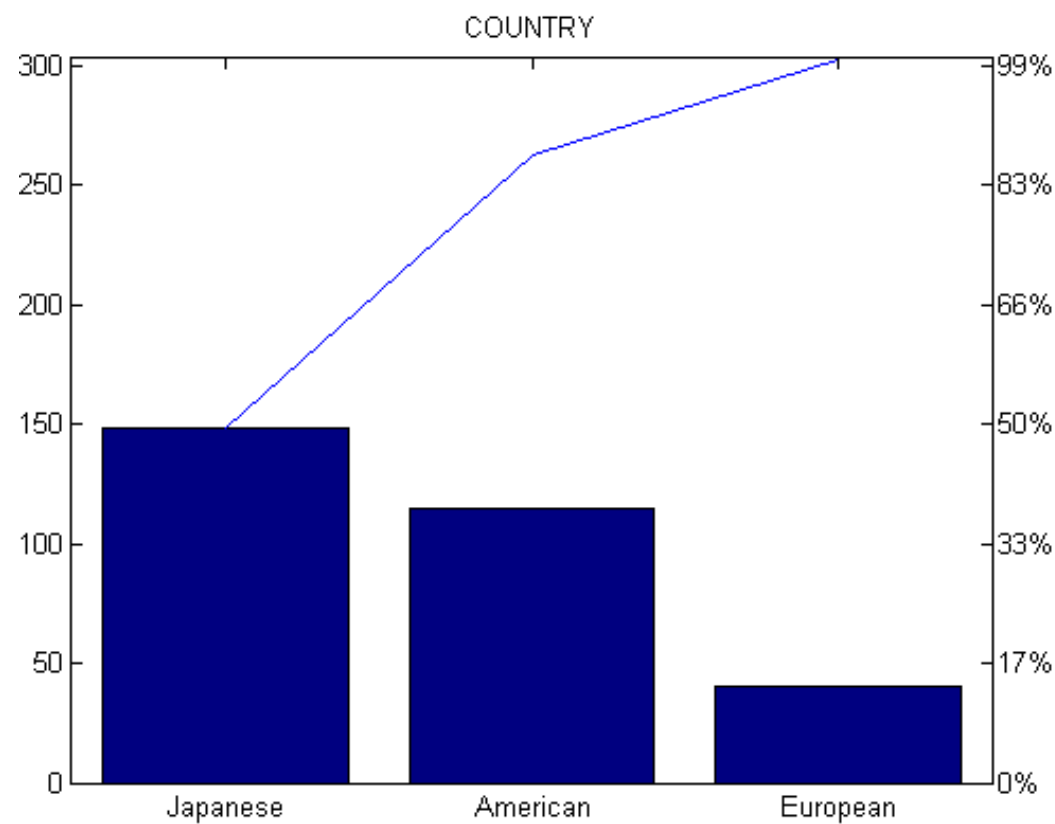
HISTOGRAM, a plot that **DISPLAYS THE DISTRIBUTION OF VALUES FOR ATTRIBUTES** by **DIVIDING THE POSSIBLE VALUES INTO BINS** and **SHOWING THE NUMBER OF RECORDS THAT FALL INTO EACH BIN**.

HISTOGRAM, for **QUANTITATIVE ATTRIBUTE** each **BIN IS AN INTERVAL OF VALUES**, bins can have the same width or not.



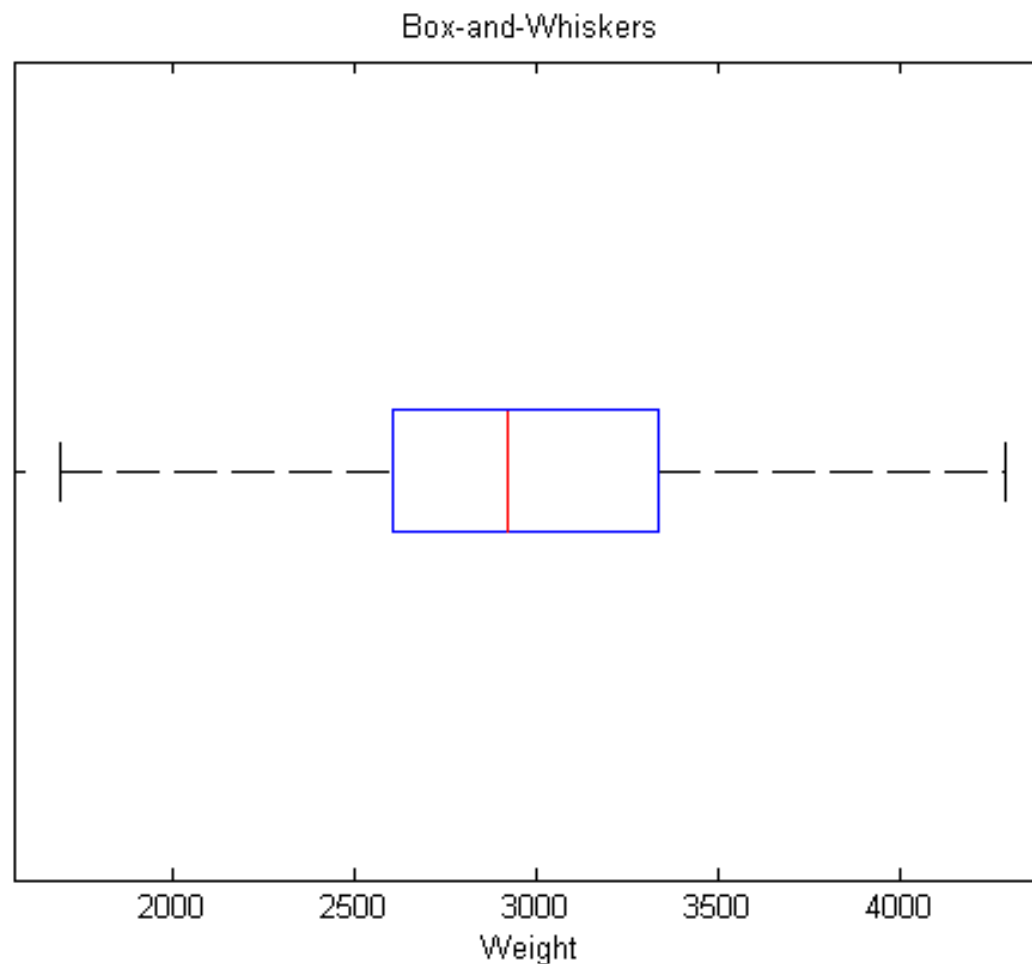
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HISTOGRAM, for **QUALITATIVE ATTRIBUTE** each **BIN IS ASSOCIATED WITH A VALUE**, when values are too much they are aggregated to possibly form meaningful bins.



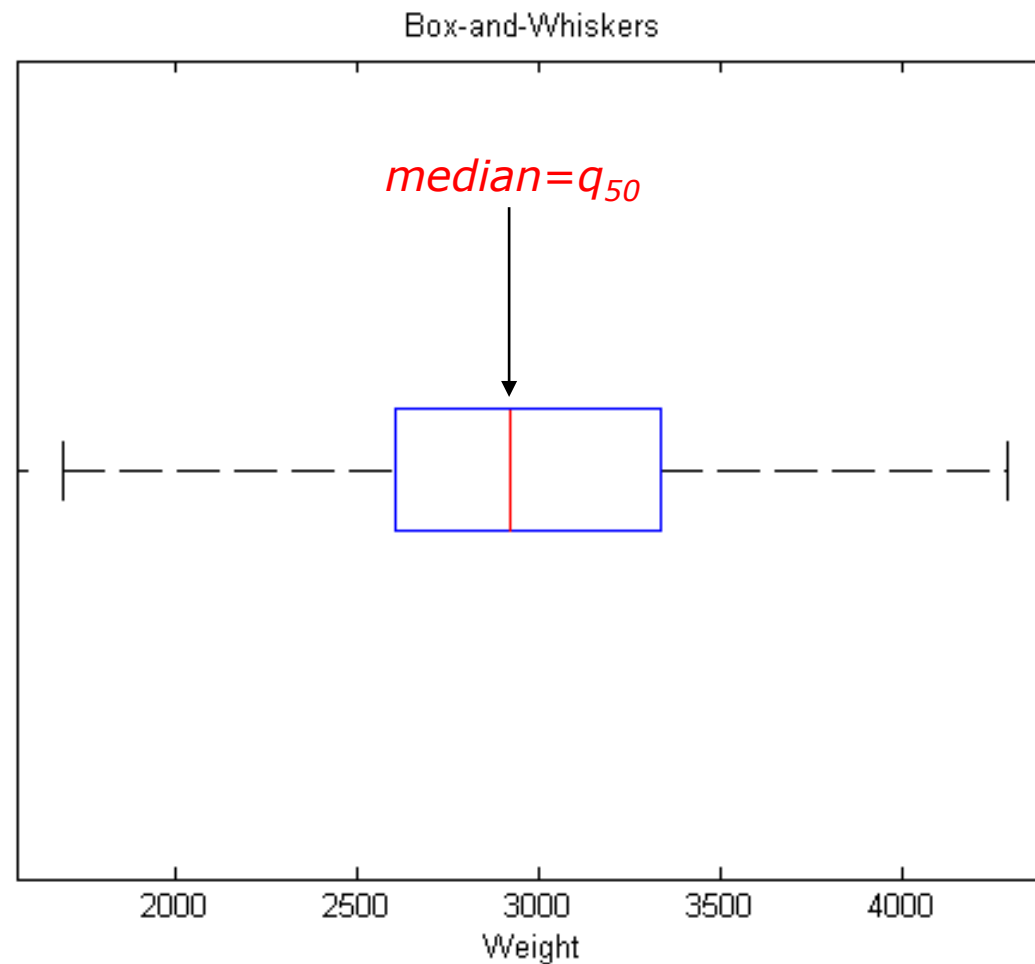
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BOX AND WHISKERS (BOX PLOT), are applied to **QUANTITATIVE ATTRIBUTES ONLY**.



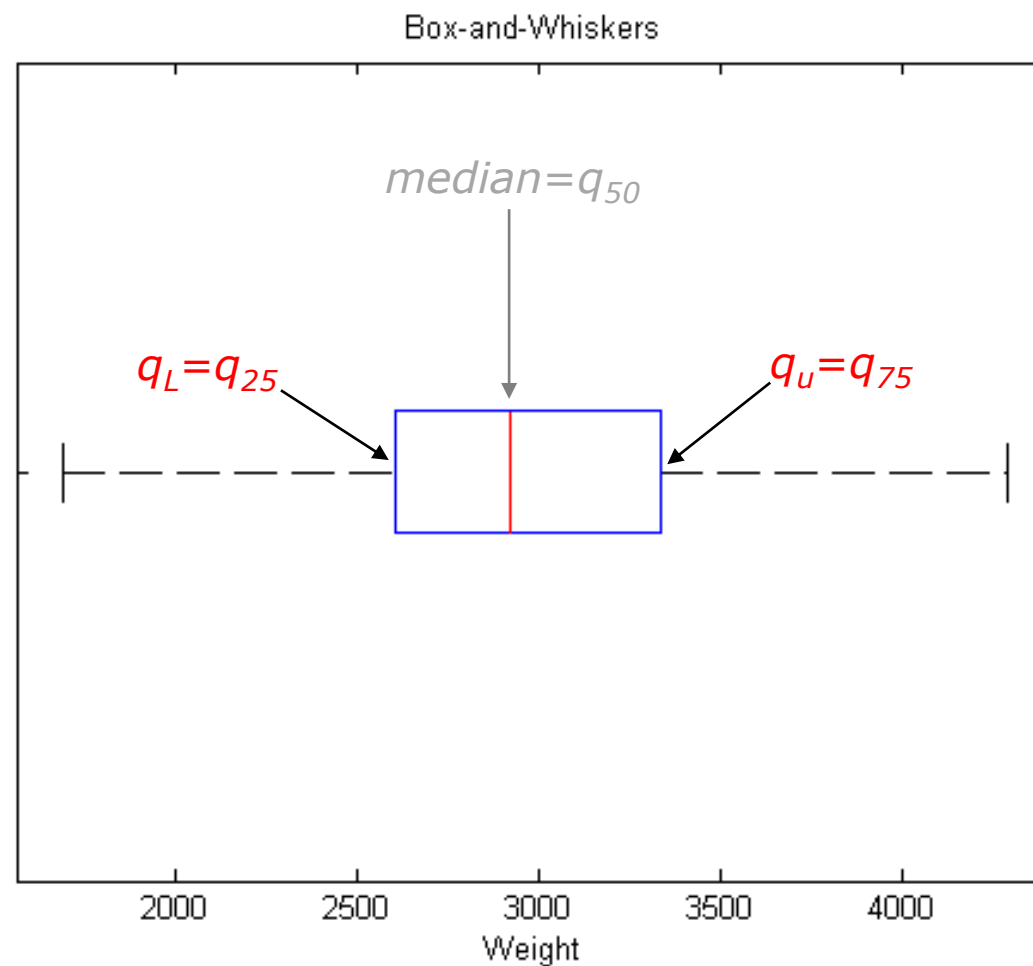
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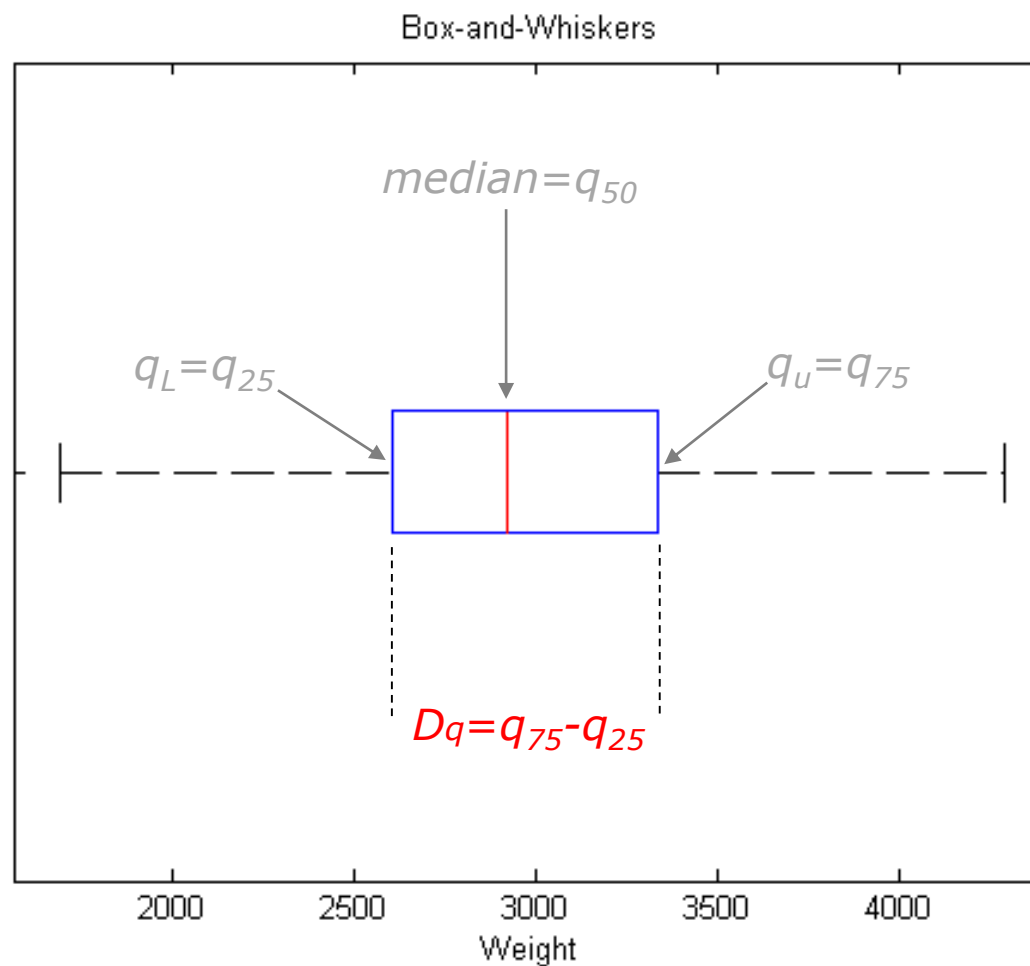
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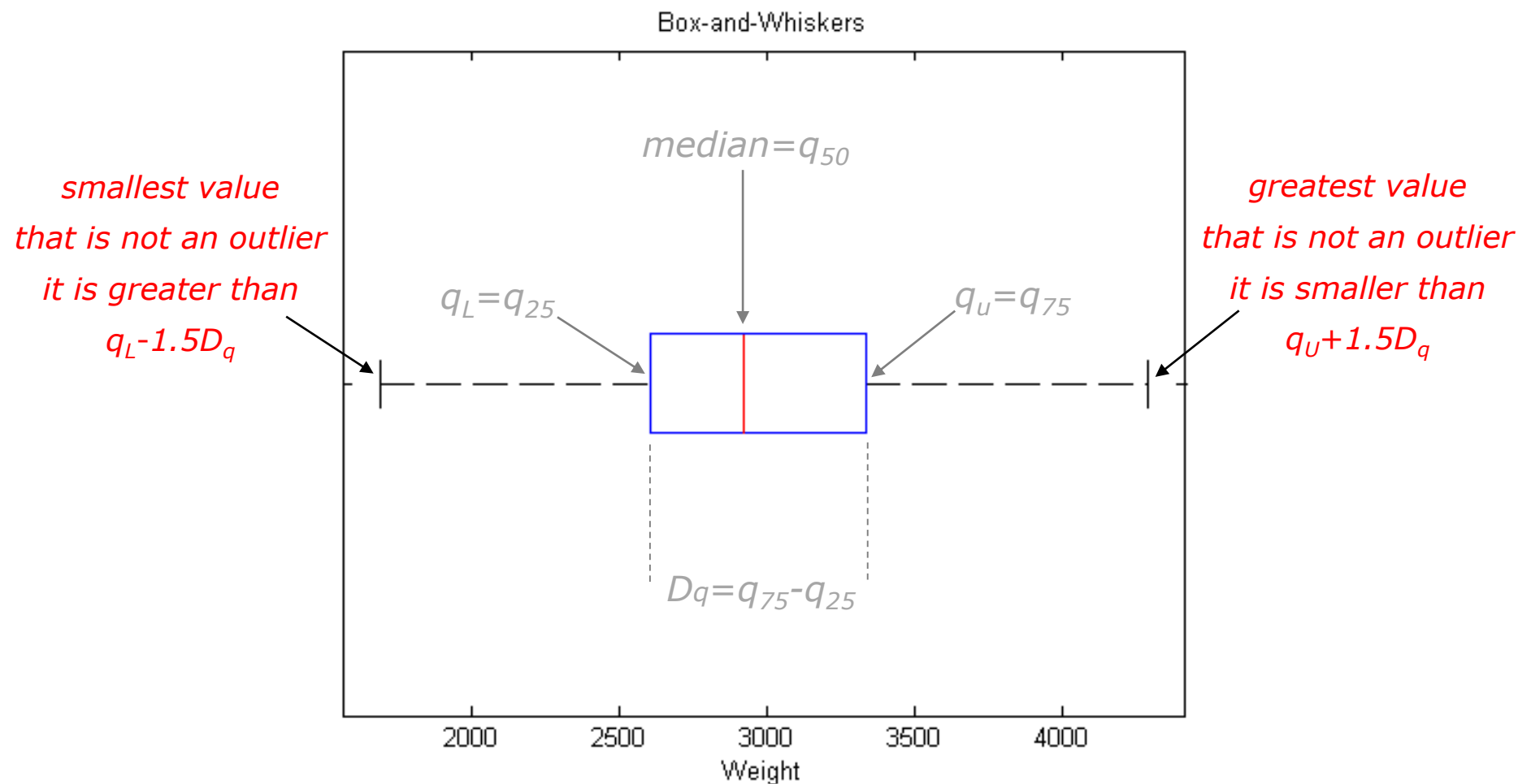
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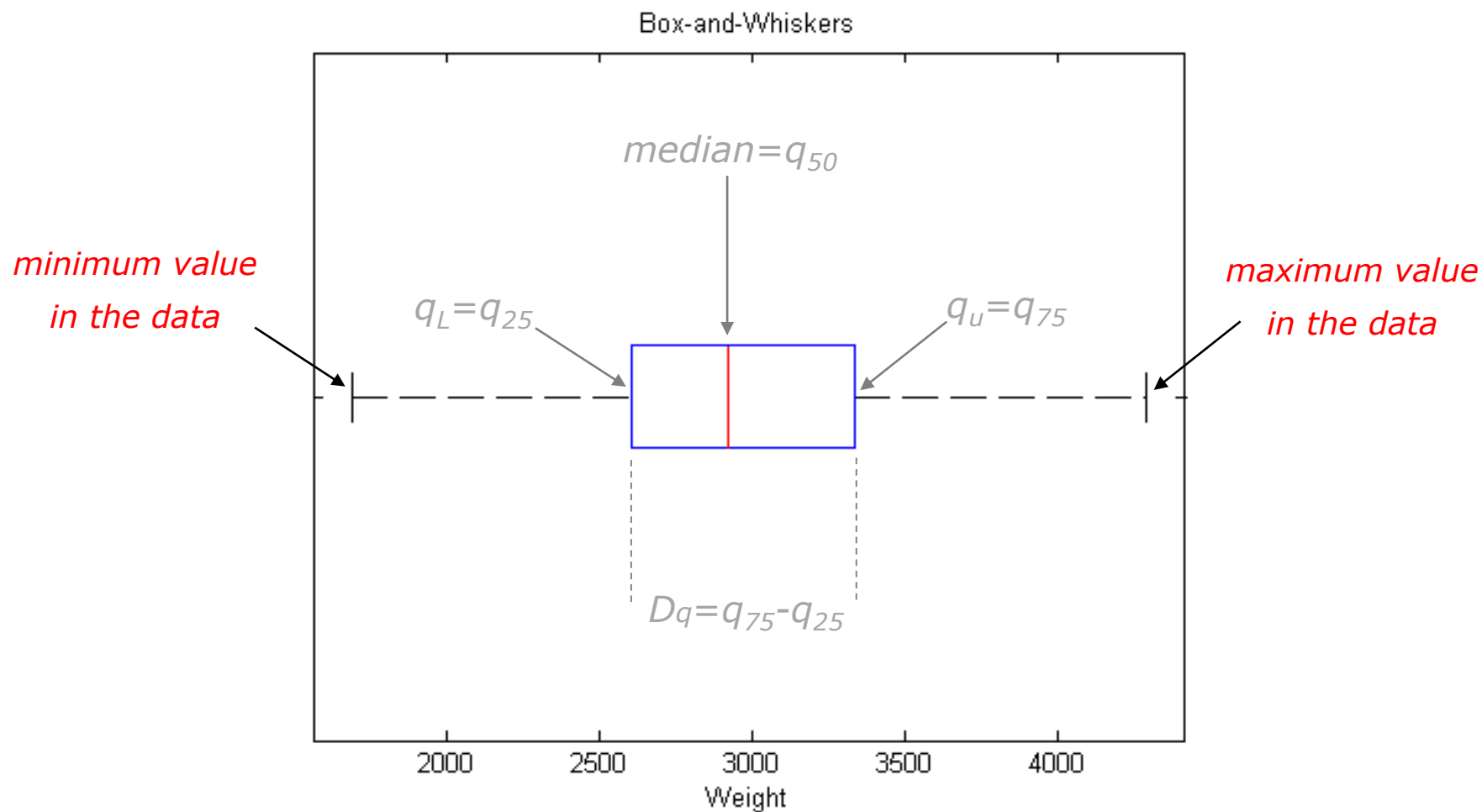
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